29-07-26

Theory

N = 6

\# $\underbrace{0101}_{ch}$ $2^4 = (0-15)$

$x_1 = 0010 = 2$
$x_2 = 1101 = 13$

x_1, x_2, x_3
x_4, x_5, x_6

fitness : $15x - x^2$
$= 15 \times 12 - (12)^2$
$= 36$

fitness ratio : $\dfrac{.x}{\sum F(x)}$

crossover :
$P_1 \rightarrow 11|01 \rightarrow 1100$
$P_2 \rightarrow 10|00 \rightarrow 1001$

0/1 Knapsack Problem

Item	Weight	value
1	3	15
2	2	8
3	4	22
4	1	6
5	2	13

Knapsack (K)
K = 7

chromosome এর population size 4 হবে
match হতে হবে,

=> Initialize $\boxed{\text{four}}$ chromosome at random, where each gene indicates whether the corresponding item is included (1) or excluded (0). For each ch. calculate the fitness prob. of selection for gen 2 Then select ch. for Gen 2 &

perform single point crossover at point 3 to generate the offspring, & show whether the resulting ch. of Gen 2 are accepted or rejected by knapsack constraint. Comment on the result

⟹ | 1 | 2 | 3 | 4 | 5 | $= 2^5 = 32$ combination

[0 থাকলে X, 1 হলে value করলো]

	1	2	3	4	5		
ch1 :	0	1	1	1	0	= 7	[accepted]
ch2 :	0	0	1	1	0	= 5<7 ;	[accepted]
ch3 :	1	1	0	1	0	6>7 ;	[accepted]
ch4 :	1	1	1	1	1	12>7 ;	rejected
	3	2	4	1	2		

2 + 4 + 1 → weight

4 + 1 → weight

number →	2	3	4
fitness value :	8	22	6

⇒ 36

Gen (1)

ch.	string	Fitness value	F. probability
e_1	01110	36	36/93 = 0.38 = 38
e_2	00110	28	28/93 = 0.30 = 30
e_3	11010	29	29/93 = 0.31 = 31
e_4	11111	0	0
		$\Sigma(x) = 93$	

condition এর সাথে unmatch

Roulette wheel distribution : new gen

e_1 : 0 1 1 1 0 → OS_1 : 011 | 10 → 01110

e_2 : 0 0 1 1 0 → OS_2 : 001 | 10 → 00110

e_2 : 0 0 1 1 0 → OS_1 : 001 | 10 → 00100

e_3 : 1 1 0 1 0 → OS_2 : 110 | 00 → 11010

Gen 2 :

ch.	string	fitness value	F. Prob
e_1'	01110 → 7 (accepted)	36	36/115 = 0.31 = 31
e_2'	00110 → 5 < 7 (")	28	28/115 = 0.24 = 24
e_3'	00100 → 4 < 7 (")	22	22/115 = 0.19 = 19
e_4'	11010 → 6 < 7 (")	29	29/115 = 0.25 = 25

$\Sigma F(x_i) = 115$ (best)

Actual	Predicted
Not spam	Not spam
spam	spam
Not spam	Not spam
spam	Not spam

1. True Positive (TP): Prediet spam and Actually spam
$\Rightarrow 3$

2. True Negative (TN): Prediet Not spam and actully
Not spam $\Rightarrow 3$

3. False positive (FP): Predicted spam but actually
Not spam $\Rightarrow 2$

4. False Negative (FN): Predicted Not spam but actually
Spam $\Longrightarrow 2$

Calculate Performance metrics

1. Accuracy: Measure the overall correctness of the model

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{3+3}{3+3+2+2} = \frac{6}{10} = 0.6$$
(60%)

2. precision (Measure the accuracy of positive predictions)

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{3}{3+2} = \frac{3}{5} = 0.6\ (60\%)$$

3. <u>Recall</u> (measure the ability to find all positive cases)

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{3}{3+2} = \frac{3}{5} = 0.6\ (60\%)$$

4. <u>F1 score</u> (The harmonic mean of precision and recall)

$$\text{F1 score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= 2 \times \frac{0.6 \times 0.6}{0.6 + 0.6}$$

$$= 0.6\ (60\%)$$

Confusion Matrix

~~used~~
Classification: → binary classification → 0 or 1
→ Multiplication "

binary classifi

T → P | F → P
T → N | F → N

[rowwise → recall
columnwise → prediction

		Predicted class: Positive	Negative	
Actual class	positive	True Positive (TP)	False Negative (FN)	sensitivity $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP)	True Negative (TN)	specificity $\frac{TN}{(TN+FP)}$
		Precision $\frac{TP}{(TP+FP)}$	Negative Predictive value $\frac{TN}{(TN+FN)}$	Accuracy $\frac{TP+TN}{(TP+TN+FP+FN)}$

Ex:

Actual	Predicted
spam	spam
Not spam	spam
spam	spam
Not spam	Not spam
Not spam	~~Snap~~ spam
spam	Not spam